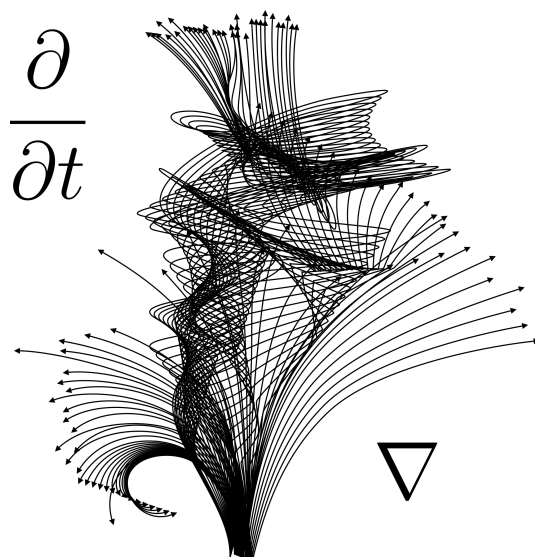




Quantitative Finance Interview Notes

Author: Xiaodong Yang CQF, FRM

Detailed information can be found in the README.md file.

Last Updated: May 9, 2026

Contents

Chapter 1	Probability, Statistics & Brainteasers	Page 1
1.1	Probability Foundations (Part A)	1
1.2	Deep Conditional Probability, Classic Puzzles & Core Interview Problems	5
1.3	Advanced Probability, Order Statistics, More Interview Questions, and More Brainteasers	13
1.4	Brownian Motion, Scaling Limits, Quadratic Variation, Hitting Times, Jensen's Inequality	13
1.5	Foundations of Estimation, Bias, Variance, MSE, and Sampling	13
1.6	Regression, OLS, HAC Standard Errors, Multicollinearity, Predictive Modeling & Interview Questions	13
1.7	Time-Series Models, Forecasting, Stationarity, Cointegration, Kalman Filter, GARCH & Interview Questions	13
1.8	Machine Learning for Quants	13
1.9	Brainteaser Compendium	13
Chapter 2	Derivatives, Greeks, Models & Stochastic Calculus	Page 14
2.1	Black-Scholes-Merton Derivations	14
2.2	Greeks	15
2.3	Trees, Monte Carlo, PDE & FEM	15
2.4	Stochastic calculus	15
2.5	Models beyond BSM	15
2.6	Hedging & replication	15
2.7	Derivatives interview Q&A	15
Chapter 3	Coding, Math, Linear Algebra & Technical Computing	Page 16
3.1	Coding Fundamentals for Quants (Python + C++)	16
3.2	Math Essentials	16
3.3	Coding + Probability	16
3.4	Practical Quant Engineering	16
3.5	Full Coding Interview Problems	16

Chapter 4	Numerical Methods, C++, & HFT-Style Algorithms	Page 17
4.1	Numerical Methods in Quant Finance	17
4.2	C++ for Quants (Low-Latency Coding)	18
4.3	High-Frequency Trading (HFT) Algorithms & Data Structures	18
4.4	Numerical Option Pricing	18
4.5	Probabilistic & Statistical Tools for HFT	18
4.6	Coding Interview Problems (C++ / Python)	18
Chapter 5	Portfolio Optimization, Execution, and Systematic Strategy Design	Page 19
5.1	Signal Construction & Alpha Modeling	20
5.2	Position Sizing & Risk Constraints	20
5.3	Portfolio Optimization (Modern Quant Version)	20
5.4	Statistical Arbitrage & Pairs Trading	20
5.5	Backtesting & Simulation Frameworks	20
5.6	Execution Algorithms & Cost Modeling	20
5.7	Performance Evaluation (Advanced)	20
5.8	Production Deployment	20
Chapter 6	Old	Page 21
6.1	Curated Question Set for Your 6-Week Plan	21
6.2	Upgraded version	30
6.3	New Problems	30

Chapter 1

Probability, Statistics & Brainteasers

1.1 Probability Foundations (Part A)

Introduction: Why Probability Dominates Quant Interviews

Probability is the single most important skill in quant interviews. A large fraction of questions - brainteasers, logic puzzles, stochastic calculus intuition, derivatives pricing, and risk-neutral reasoning - all reduce to:

- computing expectations
- understanding conditional probability
- analyzing random walks
- reasoning under uncertainty

Firms use probability questions to test your thinking speed, mathematical maturity, and structural reasoning. These problems rarely require memorized formulas; instead, they assess whether you can:

- break a problem into states
- compute conditional paths
- identify symmetry
- reason recursively
- use Markovian structure

This section builds the foundation for the more advanced topics later in the chapter.

Probability Axioms (Interview Level)

Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space.

A probability measure satisfies:

1. **Non-negativity:**
 $\mathbb{P}(A) \geq 0$
2. **Normalization:**
 $\mathbb{P}(\Omega) = 1$

3. Countable additivity:

If A_i are disjoint,

$$\mathbb{P}(\cup_i A_i) = \sum_i \mathbb{P}(A_i)$$

Interview expectation:

You never need to formally state these, but many tricky questions rely on:

- breaking events into disjoint pieces
- checking if events are mutually exclusive
- normalizing probabilities

Conditional Probability & Bayes' Rule

Definition

$$\mathbb{P}(A | B) = \frac{\mathbb{P}(A \cap B)}{\mathbb{P}(B)} \quad (1.1)$$

Bayes' rule

$$\mathbb{P}(A | B) = \frac{\mathbb{P}(B | A) \mathbb{P}(A)}{\mathbb{P}(B)} \quad (1.2)$$

Total probability

$$\mathbb{P}(B) = \sum_i \mathbb{P}(B | A_i) \mathbb{P}(A_i) \quad (1.3)$$

Interview insight: Most conditional probability questions are solved by:

1. Defining states
2. Applying total probability
3. Using Bayes' rule for inversion

Classic Interview Example

The Monty Hall Problem

Example 1.1.1

Setup: You choose 1 of 3 doors; 1 hides a prize. Host opens another door that definitely has no prize. Do you switch?

Solution (interview version):

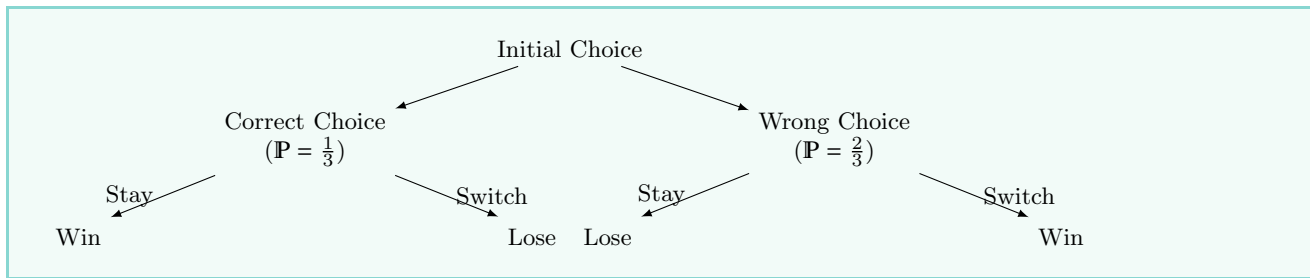
Probability chosen door is correct = $1/3$

Probability prize is behind one of the two unchosen doors = $2/3$

Host removes one losing option

Switching gives probability = $2/3$

Takeaway: If information is not independent (host has intent), conditional probability changes.



Random Variables & Common Distributions

Bernoulli Binary outcome, 0 or 1.

$$\mathbb{E}[X] = p, \quad \text{Var}(X) = p(1 - p) \quad (1.4)$$

Binomial sum of n Bernoullis.

$$\text{mean} = np, \quad \text{variance} = np(1 - p) \quad (1.5)$$

Poisson (arrivals, jumps)

$$\mathbb{P}(N_t = k) = \frac{(\lambda t)^k e^{-\lambda t}}{k!} \quad (1.6)$$

Key properties:

- Independent increments
- Stationary increments
- Mean = variance = λt

Interview use cases:

- arrival of trades
- number of jumps in jump-diffusion
- queueing logic

Normal Distribution

$$X \sim N(\mu, \sigma^2) \quad (1.7)$$

$$\mathbb{E}[X] = \mu, \quad \mathbb{E}[X^2] = \sigma^2 + \mu^2 \quad (1.8)$$

Expectations, Variance & Covariance

Linearity of expectation (critical interview tool):

$$\mathbb{E}[X + Y] = \mathbb{E}[X] + \mathbb{E}[Y] \quad (1.9)$$

Variance:

$$\text{Var}(X) = \mathbb{E}[X^2] - (\mathbb{E}[X])^2 \quad (1.10)$$

Covariance:

$$\text{Cov}(X, Y) = \mathbb{E}[XY] - \mathbb{E}[X]\mathbb{E}[Y] \quad (1.11)$$

Correlation:

$$\rho = \frac{\text{Cov}(X, Y)}{\sigma_X \sigma_Y} \quad (1.12)$$

Common Quant-Interview Expectation Tricks

Expected value of geometric trials Until first success:

$$\mathbb{E}[N] = \frac{1}{p} \quad (1.13)$$

Examples:

- die rolls until first 6 $\rightarrow 6$
- coin flips until heads $\rightarrow 2$

Expected value of repeated experiments if X_i are IID

$$\mathbb{E}\left[\frac{1}{n} \sum_{i=1}^n X_i\right] = \mathbb{E}[X] \quad (1.14)$$

Symmetry Many brainteasers are solved instantly by symmetry.

Example 1.1.2

Random chord in a circle — probability it is longer than the side of the inscribed triangle?
Answer depends on chosen definition — interviewers check if you question assumptions.

Random Walks (Discrete)

Used in:

- probability puzzles
- gambler's ruin
- Brownian motion intuition
- finance models

1D symmetric walk

$$S_{n+1} = S_n + X_n, \quad X_n = \pm 1 \quad (1.15)$$

Gambler's Ruin Probability you hit upper boundary b before lower boundary 0 starting from i : Symmetric ($p = 0.5$):

$$\mathbb{P}(\text{hit } b) = \frac{i}{b} \quad (1.16)$$

Asymmetric ($p \neq 0.5$):

$$\mathbb{P}(\text{hit } b) = \frac{1 - (q/p)^i}{1 - (q/p)^b} \quad (1.17)$$

This appears frequently in quant interviews.

Markov Chains (Interview Level)

Definition A process X_n is Markov if:

$$\mathbb{P}(X_{n+1} = x | X_n \cdots, X_0) = \mathbb{P}(X_{n+1} = x | X_n) \quad (1.18)$$

Used heavily in:

- queueing
- random walks
- credit rating transitions
- discretized stochastic models

Transition matrix

$$P = \begin{pmatrix} p_{11} & p_{12} \\ p_{21} & p_{22} \end{pmatrix} \quad (1.19)$$

n-step transitions:

$$p^n$$

Order Statistics (Interview Relevant)

if you draw n IID samples:

- $X_{(1)} = \min$
- $X_{(n)} = \max$

Example 1.1.3

$$\mathbb{E}[\max(U_1, \dots, U_n)] = \frac{n}{n+1} \quad (1.20)$$

Stopping Times (Foundation for Brownian Motion)

A stopping time τ satisfies:

$$\tau \leq t \in \mathcal{F}_t$$

Example 1.1.4

- first time a walk hits $\pm a$
- first time volatility exceeds threshold
- hitting boundary in Brownian motion (early exercise timing)

Interviews rarely ask for formal definition — they want intuition: “**A stopping time is a time at which you stop based only on current information.**”

1.2 Deep Conditional Probability, Classic Puzzles & Core Interview Problems

Conditional Probability — Core Interview Framework

Conditional probability is the single most tested concept in quant interviews. Most puzzles reduce to:

- Identify states
- Assign base probabilities

- Use Bayes' rule
- Compute conditional outcomes
- Interpret intuitively

Below are advanced forms that often confuse candidates.

The Problem of “Useful vs Useless Information”

A recurring theme is distinguishing between:

Independent information

Does not change conditional probability (e.g., rolling a fair die twice)

Dependent information Massively changes the probability (e.g., host showing a goat in Monty Hall)

Key test in interviews:

Always ask yourself: “Does this person KNOW something that influences the outcome?”

Classic Conditional Probability Puzzle Types

Type 1: Revealed information

- Monty Hall
- Cards revealed
- Doors opened
- Host behaviour

Type 2: Random selection vs conditional selection

- “You pick a random child and they are a boy...” vs
- “In a family with at least one boy...”

Type 3: Biased events

- unfair coins
- skewed dice

Type 4: Symmetry + Bayes

- random meeting problems
- birthday paradox
- dart on a square

Interview Classic Problems

Problem 1.2.1 The Two-Child Problem

A family has two children. You meet one of the children randomly, and it is a girl. What is the probability the other child is a girl?

Solution: Incorrect intuition: “Only two possibilities: GB or GG $\Rightarrow 1/2$.”

Correct reasoning: Possible equally likely families: GG, GB, BG, BB

Now you meet a random child, and it's a girl.

Compute probability of observing "Girl met" in each family:

GG family $\rightarrow$ always meet girl (prob = 1)

GB family $\rightarrow$ meet girl w.p. $1/2$

BG family $\rightarrow$ meet girl w.p. $1/2$

BB family $\rightarrow$ never

Weighted probabilities:

GG: 1×1

GB: $(1/2) \times 1$

BG: $(1/2) \times 1$

Normalize:

$$\mathbb{P}(GG|girl) = \frac{1}{1 + \frac{1}{2} + \frac{1}{2}} = \frac{1}{2}$$

Problem 1.2.2 The Drunk Passenger Problem

On a plane with n seats, the first passenger is drunk and picks a random seat. Every subsequent passenger takes their own seat if available; otherwise picks a random available seat. What is the probability the last passenger gets their own seat?

Solution:

Interview solution: This is a symmetry problem.

The last passenger's seat is taken only if:

- the drunk passenger picks their seat first
- OR someone else random-picks it later

The shocking fact:

Once the drunk picks a wrong seat, the problem resets with two "special" seats:

- seat 1
- seat n (last passenger)

Every time someone is forced to pick randomly, the choice is only between the first and last seats eventually.

Thus:

$$\mathbb{P}(\text{last person gets own seat}) = \frac{1}{2}$$

Problem 1.2.3 Random Meeting Problem

Two friends agree to meet between 0 and 1 hour. Each arrives at a random time uniformly and waits 15 minutes. What is the probability they meet?

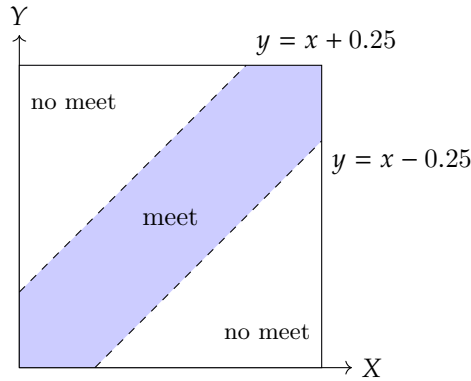
Solution: Solution via geometry

You meet if:

$$|X - Y| < 0.25$$

In the unit square, the “meeting band” is:

$$\begin{aligned}
 \text{Area} &= 1 - 2 \times (\text{triangles each with side } 0.75) \\
 &= 1 - 2 \times \frac{1}{2}(0.75)^2 \\
 &= 1 - 0.5625 \\
 &= 0.4375.
 \end{aligned}$$



Problem 1.2.4 Harder Version — Different Wait Times

Two friends agree to meet between time 0 and 1 hour.

- Friend A arrives uniformly on $[0, 1]$ and waits a hours
- Friend B arrives uniformly on $[0, 1]$ and waits b hours

(where $0 < a, b < 1$.)

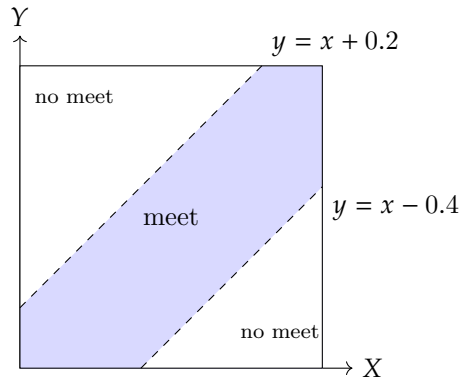
What is the probability they meet?

Solution:

Two friends arrive at times X and Y on $[0, 1]$. Friend A waits a hours and B waits b hours. They meet exactly when the arrival intervals overlap, which is $-b \leq Y - X \leq a$. For uniform arrivals this is the area of a diagonal band in the unit square. You remove two no-meeting triangles of area $\frac{1}{2}(1-a)^2$ and $\frac{1}{2}(1-b)^2$, so

$$P(\text{meet}) = 1 - \frac{(1-a)^2 + (1-b)^2}{2}.$$

For arbitrary arrival distributions you integrate the joint density over that band.



Problem 1.2.5 The Birthday Paradox

How many people are needed so that the probability of at least one shared birthday is 50%?

Solution:

$$\mathbb{P}(\text{no match}) = \frac{365}{365} \times \frac{364}{365} \times \cdots \times \frac{365 - n + 1}{365}$$

Set equal to $\frac{1}{2}$, solve numerically.

Problem 1.2.6 The Balls & Bins Problem

Throw n balls into n bins independently and uniformly. What is the expected number of empty bins?

Solution: For a given bin:

$$\mathbb{P}(\text{empty}) = \left(1 - \frac{1}{n}\right)^n \approx e^{-1}$$

So:

$$\mathbb{E}[\text{empty bins}] = ne^{-1}$$

Deep Dive: Gamma(Shape, Scale)

Gamma distributions show up in:

- waiting times
- variance-gamma models (used in quant finance)
- Bayesian statistics
- mixture models

Properties

$$X \sim \Gamma(k, \theta)$$

$$\mathbb{E}[X] = k\theta, \quad \text{Var}(X) = k\theta^2$$

Interviews often test “identify the right distribution” when:

- time between jumps
- sum of exponential variables

Quant Interview Questions

Problem 1.2.7

Roll a die repeatedly. What is the expected number of rolls until you see the sequence 1-1 (two ones in a row)?

Solution:

This is a Markov chain with 2 states:

S_0 = no previous 1

S_1 = last roll was 1

Solve linear equations: Expected rolls = $36/11 \approx 3.27$

Problem 1.2.8

What is the probability that the maximum of two independent Uniform(0,1) variables exceeds 0.8?

Solution:

$$1 - 0.8^2 = 0.36$$

Problem 1.2.9

A fair coin is tossed until heads. Expected tosses?

Problem 1.2.10

A box has 3 white and 2 black balls. Two are drawn without replacement. Probability both white?

Problem 1.2.11

What is the expected value of the maximum of 3 IID Uniform(0,1) variables?

Problem 1.2.12

What is $\mathbb{E}[X|X > 1]$ for exponential rate 1?

Problem 1.2.13

Draw a random point in a unit square. Probability $x + y < 1$?

Problem 1.2.14

Flip a fair coin 10 times. Probability of exactly 5 heads?

Problem 1.2.15

If two stocks have correlation ρ , what is the covariance?

Problem 1.2.16

Probability that Poisson(λ) variable is even?

Brainteasers

Brainteasers 1.2.1

The Light Switch Puzzle (Short Version)

Solution:

Turn switch A on for 5 minutes

Turn off A, turn on B

Enter room:

Light on $\rightarrow$ B

Light off but warm $\rightarrow$ A

Light off and cold $\rightarrow$ C

Brainteasers 1.2.2

Two Ropes Burning (45 Minutes)

Solution:

Burn Rope 1 at both ends

Burn Rope 2 at one end

When Rope 1 finishes (30 min), light Rope 2 other end
Rope 2 burns remaining 15 min
Total = 45 min

Brainteasers 1.2.3

100 Lockers

Solution:

Only lockers with square numbers stay open (because toggled odd number of times).

Brainteasers 1.2.4

Find Heavier Ball With Two Weighings

Solution:

Use a $3 \times 3 \times 3$ splitting; identify remainder group recursively.

Brainteasers 1.2.5

Farmer, Wolf, Sheep, Cabbage

Solution: Sheep first, return alone, wolf next, bring back sheep, cabbage over, return empty, sheep last.

Brainteasers 1.2.6

Mislabeled Bags

Solution: Draw one fruit from the “mixed” label (which must be wrong). Deduce all labels.

Brainteasers 1.2.7

Ants on a Stick

Solution: Expected time until fall depends on boundaries and speeds— but direction doesn’t matter: ants can be treated as ghosts passing through.

Brainteasers 1.2.8

12 Coins, One Fake (Unknown Heavier/Light)

Solution: Three weighings $\rightarrow$ ternary decision tree ($27 \text{ leaves} \geq 24 \text{ possibilities}$).

Brainteasers 1.2.9

The Hotel Bill Problem

Solution: (paradox with missing \$1; resolve misdirection)

Brainteasers 1.2.10

Monty Hall Variation

Solution: Stay vs switch when multiple goats opened $\rightarrow$ generalize to $(n - 1)/n$ win rate when switching.

1.3 Advanced Probability, Order Statistics, More Interview Questions, and More Brainteasers

Advanced Probability: Decomposition, Law of Total Expectation

Advanced Bayes Rule Use-Cases in Interviews

Order Statistics (Deep Dive)

Extreme Value Behavior (Useful for your actual PhD research)

Advanced Interview Questions (20 More)

Brainteasers (10 More, With Solutions)

Advanced Distributions for Quant Interviews + Heavy Tails + Extreme Value Theory

Lognormal Distribution - Core of Quant Finance

Student-t Distribution - Why It Matters for Quant Jobs

Laplace Distribution - The “Double Exponential”

Mixture Models (Gaussian Mixtures)

Extreme Value Theory (EVT) - Connection to My Research

Maximum Domain of Attraction - When Extremes Behave Nicely

Advanced: Lognormal Mixture = Heavy Tail

20 Advanced Probability Interview Questions (With Solutions)

10 Advanced Brainteasers

1.4 Brownian Motion, Scaling Limits, Quadratic Variation, Hitting Times, Jensen’s Inequality

From Random Walk to Brownian Motion (Central Insight)

Definition of Brownian Motion (Interview Level)

Crucial Property: Quadratic Variation

Stopping Times - Deepening the Intuition

Hitting Times of Brownian Motion

Reflection Principle (Very Common in Interviews)

Scaling Property of Brownian Motion

Jensen’s Inequality (Very Important) 13

Optional Stopping Theorem (Intuition Only)

10 Bridge to Stochastic Calculus (Setting Up Chapter 2)

Chapter 2

Derivatives, Greeks, Models & Stochastic Calculus

2.1 Black-Scholes-Merton Derivations

This section is one of the MOST important parts of quantitative finance.
Top firms LOVE asking:

- “Derive Black-Scholes.”
- “Why does μ disappear?”
- “Explain risk-neutral pricing.”
- “Explain delta hedging.”
- “What is replication?”
- “What is the intuition behind Girsanov?”

You must know multiple derivations because different interviewers prefer different perspectives.

Why Does Risk-Neutral Pricing Work?

This is a COMMON interview question.

Key Idea

We are NOT saying investors are risk-neutral.
Instead:

- hedging removes risk
- therefore derivative price must equal replication cost
- under no-arbitrage, there exists equivalent martingale measure $\mathbb{Q}$

Under $\mathbb{Q}$: all discounted tradable assets are martingales.

Derivation 4: Feynman-Kac Representation

The PDE:

$$V_t + \mathcal{L}V - rV = 0$$

with terminal payoff:

$$V(S, T) = \Phi(S)$$

has solution:

$$V(S, t) = E^{\mathbb{Q}}[e^{-r(T-t)}\Phi(S_T)]$$

This is the Feynman-Kac theorem.

It connects:

- PDEs
- stochastic calculus
- expectations
- pricing theory

This is VERY important for advanced quant interviews.

2.2 Greeks

2.3 Trees, Monte Carlo, PDE & FEM

2.4 Stochastic calculus

2.5 Models beyond BSM

2.6 Hedging & replication

2.7 Derivatives interview Q&A

Chapter 3

Coding, Math, Linear Algebra & Technical Computing

3.1 Coding Fundamentals for Quants (Python + C++)

3.2 Math Essentials

3.3 Coding + Probability

3.4 Practical Quant Engineering

3.5 Full Coding Interview Problems

Chapter 4

Numerical Methods, C++, & HFT-Style Algorithms

4.1 Numerical Methods in Quant Finance

Root Finding

TextTextTextTextTextTextTextTextTextTextTextTextTextTextTextText

Numerical Integration

Finite Difference Methods for PDEs

Monte Carlo Methods (Advanced)

Numerical Linear Algebra

4.2 C++ for Quants (Low-Latency Coding)

C++ Basics for Speed

Memory & Cache Optimization

Efficient Data Structures

Exception Safety, Error Handling

Build optimization (O2/O3, LTO, PGO)

4.3 High-Frequency Trading (HFT) Algorithms & Data Structures

Order Book Modeling

Microstructure Models

HFT Data Structures

Latency-Sensitive Coding

Simulating Order Books

4.4 Numerical Option Pricing

Black-Scholes (Analytic and Numeric)

Binomial & Trinomial Trees

PDE Methods

Heston & SV Models

Jumps (Merton, Bates)

4.5 Probabilistic & Statistical Tools for HFT

Poisson & Hawkes Processes

Microprice & Queue Estimation

Market Impact Models

4.6 Coding Interview Problems (C++ / Python)

Chapter 5

Portfolio Optimization, Execution, and Systematic Strategy Design

5.1 Signal Construction & Alpha Modeling

Types of signals

Forecasting returns

Combining forecasts

5.2 Position Sizing & Risk Constraints

5.3 Portfolio Optimization (Modern Quant Version)

Classical MPT failure modes

Robust optimization

Shrinkage covariance matrices

Black-Litterman

Hierarchical Risk Parity (HRP)

Convex optimization

5.4 Statistical Arbitrage & Pairs Trading

Mean-reversion signals

Z-score normalization

Kalman-filter pairs (time-varying beta)

Cointegration

Spread modeling

Execution of stat-arb portfolios

5.5 Backtesting & Simulation Frameworks

Chapter 6

Old

6.1 Curated Question Set for Your 6-Week Plan

Week 1 - Probability Foundations

Focus: conditional probability, expectations, variance, random walks.

Problem 6.1.1

A biased coin has probability p of heads. How many flips do you expect to need to get two consecutive heads?

Solution:

We define the following states based on the current progress toward the goal of "HH" (two consecutive heads): State 0 (S_0): We have not yet flipped the coin, or the last flip was a Tail. We have zero consecutive heads. State 1 (S_1): Our last flip was a Head. We need one more Head to reach the goal. State 2 (S_2): We have achieved two consecutive heads. This is our stopping state.

Let E_0 be the expected number of additional flips needed to reach S_2 starting from S_0 , and E_1 be the expected number of additional flips needed starting from S_1 . We want to find E_0 .

From State 0: On the next flip: With probability p , we get a Head and move to S_1 . With probability $q = 1 - p$, we get a Tail and stay in S_0 . The equation is:

$$E_0 = 1 + pE_1 + qE_0$$

From State 1: On the next flip: With probability p , we get a Head and move to S_2 (the goal). Since we have reached the goal, the additional flips needed is 0. With probability $q = 1 - p$, we get a Tail and move back to S_0 . The equation is:

$$E_1 = 1 + p(0) + qE_0 = 1 + qE_0$$

3. Solving the System We substitute the expression for E_1 into the equation for E_0 :

$$E_0 = 1 + p(1 + qE_0) + qE_0$$

$$E_0 = 1 + p + pqE_0 + qE_0$$

Now, group the E_0 terms on one side:

$$E_0 - pqE_0 - qE_0 = 1 + p$$

$$E_0(1 - q(p + 1)) = 1 + p$$

Substitute $q = 1 - p$ into the term $(1 - q(p + 1))$:

$$1 - (1 - p)(1 + p) = 1 - (1 - p^2) = p^2$$

So the equation becomes:

$$E_0(p^2) = 1 + p$$

$$E_0 = \frac{1+p}{p^2}$$

Final Answer The expected number of flips to get two consecutive heads is:

$$E = \frac{1+p}{p^2}$$

Example 6.1.1 (Verification with Common Values)

The expected number of flips to get two consecutive heads is:

$$E = \frac{1+p}{p^2}$$

Verification with Common Values Fair Coin ($p = 1/2$):

$$E = \frac{1+0.5}{0.5^2} = \frac{1.5}{0.25} = 6 \text{ flips}$$

Certain Heads ($p = 1$):

$$E = \frac{1+1}{1^2} = 2 \text{ flips}$$

(As expected, since every flip is a head, we get HH in exactly 2 flips).

Problem 6.1.2

A box has red, green, and blue balls. You draw two without replacement. What is the probability the colors match?

Solution:

To solve this, we need to look at the total number of ways to pick two balls and how many of those pairs result in a match. Because we aren't given the specific quantities of each color, let's define them:

- r = Number of red balls
- g = Number of green balls
- b = Number of blue balls
- $n = r + g + b$ (Total number of balls)

The total number of ways to choose 2 balls out of n without replacement (where order doesn't matter) is given by the combination formula:

$$\binom{n}{2} = \frac{n(n-1)}{2}$$

For the colors to match, you must draw either two red balls, two green balls, or two blue balls. We calculate the combinations for each:

- Two Red: $\binom{r}{2} = \frac{r(r-1)}{2}$
- Two Green: $\binom{g}{2} = \frac{g(g-1)}{2}$
- Two Blue: $\binom{b}{2} = \frac{b(b-1)}{2}$

The total number of favorable outcomes is the sum of these three:

$$\text{Favorable} = \frac{r(r-1) + g(g-1) + b(b-1)}{2}$$

The probability $P(\text{Match})$ is the ratio of favorable outcomes to total outcomes:

$$P(\text{Match}) = \frac{\frac{r(r-1)+g(g-1)+b(b-1)}{2}}{\frac{n(n-1)}{2}}$$

By canceling out the 2 in the denominators, we get the final formula:

$$P(\text{Match}) = \frac{r(r-1) + g(g-1) + b(b-1)}{n(n-1)}$$

You can also think of this as the sum of the probabilities of picking two of each specific color:

- $P(\text{Red and Red}): \frac{r}{n} \cdot \frac{r-1}{n-1}$
- $P(\text{Green and Green}): \frac{g}{n} \cdot \frac{g-1}{n-1}$
- $P(\text{Blue and Blue}): \frac{b}{n} \cdot \frac{b-1}{n-1}$

Adding these together gives the same result:

$$P(\text{Match}) = \frac{r(r-1) + g(g-1) + b(b-1)}{n(n-1)}$$

Note: If any color has fewer than 2 balls (e.g., $r = 1$), the term for that color becomes 0, correctly reflecting that it's impossible to draw a matching pair of that color.

Problem 6.1.3

A random variable X is normally distributed. What is the distribution of X^2 ?

Solution:

The distribution of X^2 depends entirely on the parameters of the original normal distribution, specifically the mean (μ) and the variance (σ^2).

1. The Special Case: Standard Normal Distribution If X is a Standard Normal random variable, meaning $X \sim N(0, 1)$, then X^2 follows a Chi-squared distribution with 1 degree of freedom.

$$X^2 \sim \chi_1^2$$

This is the fundamental definition of the chi-squared distribution.

2. The Case: $X \sim N(0, \sigma^2)$ If the mean is zero but the variance is not 1, we can normalize the variable. Since $(X/\sigma) \sim N(0, 1)$, then:

$$(X/\sigma)^2 = \frac{X^2}{\sigma^2} \sim \chi_1^2$$

In this case, X^2 follows a Scaled Chi-squared distribution, specifically $\sigma^2 \chi_1^2$, which is also a specific case of the Gamma distribution:

$$X^2 \sim \text{Gamma}\left(k = \frac{1}{2}, \theta = 2\sigma^2\right)$$

3. The General Case: $X \sim N(\mu, \sigma^2)$ If the mean μ is non-zero, the distribution of X^2 becomes more complex. Non-central Chi-squared Distribution: First, we scale the variable to have unit variance: $Z = X/\sigma$. Then $Z \sim N(\mu/\sigma, 1)$. The square of this variable, $(X/\sigma)^2$, follows a Non-central Chi-squared distribution with 1 degree of freedom and non-centrality parameter $\lambda = (\mu/\sigma)^2$. Therefore, for the general case, X^2 is a scaled version of a non-central chi-squared variable:

$$X^2 \sim \sigma^2 \chi_1^2 \left(\lambda = \frac{\mu^2}{\sigma^2} \right)$$

Problem 6.1.4

A stock price follows a random walk with upward probability 0.55. What is the probability it hits +5 before -5?

Solution:

This is a classic variation of the Gambler's Ruin problem. We can model the stock price movement as a sequence of independent steps where the price increases by 1 with probability p and decreases by 1 with probability $q = 1 - p$.

1. Variables and Parameters In this scenario: $p = 0.55$ (Probability of moving up) $q = 1 - 0.55 = 0.45$ (Probability of moving down) $s = 0$ (Starting position) $a = -5$ (Lower boundary) $b = +5$ (Upper boundary) We want to find the probability of reaching b before a . To simplify the math, we can shift the coordinates so the process starts at $i = 5$, with a lower bound at 0 and an upper bound at $N = 10$.

2. The Gambler's Ruin Formula For a non-symmetric random walk ($p \neq q$), the probability P_i of reaching the upper boundary N starting from position i is given by:

$$P_i = \frac{1 - (q/p)^i}{1 - (q/p)^N}$$

3. Calculation Let's plug in our specific values: The ratio $\frac{q}{p} = \frac{0.45}{0.55} = \frac{9}{11} \approx 0.818$ Starting position $i = 5$ (relative to the lower bound) Total distance $N = 10$

$$P_5 = \frac{1 - (9/11)^5}{1 - (9/11)^{10}}$$

Now, calculating the powers: $(9/11)^5 \approx 0.368$ $(9/11)^{10} \approx (0.368)^2 \approx 0.135$ Substituting these back into the equation:

$$P_5 \approx \frac{1 - 0.368}{1 - 0.135} = \frac{0.632}{0.865}$$

$$P_5 \approx 0.7306$$

Final Result The probability that the stock hits +5 before it hits -5 is approximately 0.731 (or 73.1%). Even though the upward bias (0.55) seems small, the "drift" significantly stacks the odds in favor of hitting the upper target over a 10-step total range.

Problem 6.1.5

Roll a fair die until you get a 6. What is the expected number of rolls?

Solution:

This is a classic problem involving a Geometric Distribution. We are looking for the expected number of trials until the first "success."

1. Defining the Probability For a fair six-sided die, the probability of rolling a 6 (success) on any given roll is:

$$p = \frac{1}{6}$$

The probability of rolling anything else (failure) is:

$$q = 1 - p = \frac{5}{6}$$

2. The Expected Value Formula For a geometric distribution, the expected value E (the average number of trials to get the first success) is given by the reciprocal of the probability of success:

$$E = \frac{1}{p}$$

3. Calculation Plugging in our value for p :

$$E = \frac{1}{1/6} = 6$$

Why is it $1/p$? (The Intuitive Proof)

Proof. If you aren't familiar with the formula, you can derive it using a simple recursive logic similar to the "consecutive heads" problem: Let E be the expected number of rolls. On the first roll, you get a 6 with probability $1/6$. If this happens, you are done (1 roll). You fail to get a 6 with probability $5/6$. If this happens, you have "wasted" one roll and are essentially back at the beginning. You still expect to need E more rolls. The equation looks like this:

$$E = \frac{1}{6}(1) + \frac{5}{6}(1 + E)$$

$$E = \frac{1}{6} + \frac{5}{6} + \frac{5}{6}E$$

$$E = 1 + \frac{5}{6}E$$

Subtract $\frac{5}{6}E$ from both sides:

$$\frac{1}{6}E = 1$$

$$E = 6$$

☺

Final Result You expect to need 6 rolls to get a 6.

It makes intuitive sense: if something happens 1 out of every 6 times on average, you'd expect to wait about 6 times for it to occur!

Problem 6.1.6

You flip a fair coin 10 times. What's the probability of exactly 5 heads?

Solution:

Problem 6.1.7

if $X \sim \text{Poisson}(\lambda)$, compute $\text{Var}(X)$ and explain why.

Solution:

To find the probability of exactly 5 heads in 10 flips of a fair coin, we use the Binomial Distribution.

1. Identify the Parameters For a binomial problem, we need four values: $n = 10$: The total number of trials (flips). $k = 5$: The number of successful outcomes (heads) we want. $p = 0.5$: The probability of success on a single trial. $q = 1 - p = 0.5$: The probability of failure (tails).

2. The Binomial Formula The probability of getting exactly k successes in n trials is:

$$P(X = k) = \binom{n}{k} p^k q^{n-k}$$

3. Step-by-Step Calculation

Step A: Calculate the Combinations First, we find how many different ways we can arrange 5 heads in 10 flips. This is "10 choose 5":

$$\binom{10}{5} = \frac{10!}{5!(10-5)!} = \frac{10 \times 9 \times 8 \times 7 \times 6}{5 \times 4 \times 3 \times 2 \times 1} = 252$$

Step B: Calculate the Probabilities Since p and q are both 0.5, the probability part of the formula simplifies:

$$p^5 \times q^5 = (0.5)^5 \times (0.5)^5 = (0.5)^{10}$$

$$(0.5)^{10} = \frac{1}{1024} \approx 0.0009765625$$

Step C: Multiply Together

$$P(X = 5) = 252 \times \frac{1}{1024}$$

$$P(X = 5) = \frac{252}{1024} = \frac{63}{256}$$

Final Result The probability of getting exactly 5 heads in 10 flips is $\frac{63}{256}$, which is approximately 0.2461 or 24.61%.

Why isn't it 50%?

It's a common intuition to think that because heads and tails are equally likely, you should get exactly 5 heads half the time. However, 5 is simply the most likely outcome (the mode); there are many other possibilities (like 4 or 6 heads) that split the remaining probability.

Problem 6.1.8

You observe one normal variable. How do you estimate its variance

Solution:

To estimate the variance of a normal distribution based on a single observation ($n = 1$), we run into a fundamental mathematical wall: you cannot calculate a sample variance from one data point. However, how you approach this depends on whether you are looking for a standard statistical estimate or a Bayesian inference.

1. The Frequentist Reality: It's Undefined In classical statistics, the formula for sample variance (s^2) requires at least two observations because it measures the "spread" or deviation from a mean.

$$s^2 = \frac{\sum (x_i - \bar{x})^2}{n - 1}$$

If $n = 1$, the denominator becomes $1 - 1 = 0$. This leads to an undefined result. Intuitively, this makes sense: if you only see one person's height (e.g., 5'10"), you have no way of knowing if everyone else is also exactly 5'10" (zero variance) or if heights range from 4' to 7' (high variance).

2. The Maximum Likelihood Approach If you attempt to use Maximum Likelihood Estimation (MLE), which uses n instead of $n - 1$, the formula is:

$$\hat{\sigma}^2 = \frac{(x_1 - \mu)^2}{1}$$

If the mean (μ) is known: You can estimate the variance as the squared deviation from that known mean. If the mean (μ) is unknown: The MLE for the mean is the observation itself (x_1). This results in a variance estimate of $(x_1 - x_1)^2 = 0$. This is a "biased" estimate and almost certainly incorrect, as it assumes the population has no spread.

3. The Bayesian Approach (The "Workaround") If you absolutely must have an estimate from one data point, you use Prior Knowledge. You don't start from zero; you start with an assumption about what the variance might be (a "Prior Distribution").

Prior: You assume the variance follows an Inverse-Gamma distribution (the conjugate prior for variance).

Observation: You see x_1 .

Posterior: You update your belief. The single observation "pulls" the prior toward the value of $(x_1 - \mu)^2$.

The Bottom Line: With $n = 1$, you can't measure "difference" because there is nothing to differ from. You either need more data, a known mean, or a strong prior assumption.

WEEK 2 — Brainteasers & Logic

From Chapter 8 (rewritten)

Problem 6.1.9

You have 3 light switches and 1 bulb in a separate room. How do you determine which switch controls the bulb with one visit?

Solution: ss

Problem 6.1.10

You have two ropes. Each burns in exactly one hour but not at a constant rate. How do you measure 45 minutes?

Problem 6.1.11

You stand in front of 100 closed lockers. You toggle every k -th locker on pass k . Which lockers stay open?

Problem 6.1.12

A man pushes his car to a hotel and is immediately bankrupt. Explain.

Problem 6.1.13

You are given 8 balls; one is heavier. How do you find the heavy one using two weighings?

Problem 6.1.14

You need to divide 8 liters into 2 and 6 using only a 5-L and 3-L container.

Problem 6.1.15

You and another person each draw a random real number in $(0,1)$. You choose separately whether to select or pass. Strategy?

WEEK 3 — Options & Derivatives

Based on Chapters 2.2.1, 2.2.2, 2.2.3, 2.2.4

Problem 6.1.16

Re-derive the Black-Scholes PDE using delta hedging.

Problem 6.1.17

Explain intuitively why Gamma is always positive for calls and puts.

Problem 6.1.18

Why does Vega always decrease as time to maturity approaches zero?

Problem 6.1.19

Construct a replicating portfolio for a binary option.

Problem 6.1.20

Explain why call prices increase with volatility.

Problem 6.1.21

How can you price an Asian option with Monte Carlo?

Problem 6.1.22

Why is implied volatility skewed in equity index options?

Problem 6.1.23

Explain the intuition behind local volatility vs stochastic volatility.

Problem 6.1.24

Show how to approximate Delta and Gamma numerically.

WEEK 4 — Math + Stochastic Processes

Based on Chapters 3 & 6

Problem 6.1.25

State and apply Itô's lemma to $X_t = S_t^2$.

Problem 6.1.26

A Brownian motion hits level 1. What is the distribution of the hitting time?

Problem 6.1.27

Solve the SDE $dX_t = \mu X_t dt + \sigma dW_t$.

Problem 6.1.28

Show that GBM is lognormal.

Problem 6.1.29

Compute $\mathbb{E}[\int_0^T W_t dt]$ and $\text{Var}[\int_0^T W_t dt]$.

Problem 6.1.30

If two Brownian motions have correlation ρ , write a simulation scheme.

Problem 6.1.31

Does a Poisson process have independent increments? Explain.

Problem 6.1.32

Interpret the drift and volatility of the OU process in simple language.

WEEK 5 — Coding & Numerical Methods

Based on Chapters 5 & 7

Problem 6.1.33

Write a Monte Carlo simulation for GBM.

Problem 6.1.34

Implement a function to compute the sample covariance matrix.

Problem 6.1.35

Implement finite-difference approximation of Delta.

Problem 6.1.36

Price a European option using variance reduction (Antithetic).

Problem 6.1.37

Simulate a correlated Brownian motion vector.

Problem 6.1.38

Implement a simple trinomial tree.

Problem 6.1.39

Price a digital call with Monte Carlo + control variates.

Problem 6.1.40

Compute PCA of a covariance matrix using NumPy.

WEEK 6 — Mock Interviews + Soft Questions

Based on Chapters 1 & 9

Technical questions

Problem 6.1.41

Explain your research in 30 seconds.

Problem 6.1.42

What is tail risk?

Problem 6.1.43

What is the intuition behind volatility risk premium?

Problem 6.1.44

How does implied skew relate to crash risk?

Problem 6.1.45

Why are index options negatively skewed?

Problem 6.1.46

Compare your LTI measure to existing tail measures (BT11Q).

Soft questions**Problem 6.1.47**

Describe a time you had to debug a complex model.

Problem 6.1.48

How would your supervisor describe you?

Problem 6.1.49

What is your biggest weakness in quant finance?

6.2 Upgraded version

6.3 New Problems

Deep Conditional Probability, Classic Puzzles & Core Interview Problems

Quant Interview Questions